

Neha Sengar

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RESEARCH INTERESTS

Mechanistic interpretability of large vision-language models, training-free visual grounding, AI-driven methods for robot perception & 3D scene understanding, open-vocabulary scene reasoning, real-world deployment in autonomous systems, generalization.

EDUCATION

Ph.D. in Robotics Program

Feb 2023 – Jan 2027 (expected)

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea

- Advisor: Prof. Dongsoo Har
- Dissertation: “*Mechanistic Interpretability of Vision-Language Models for Language-Guided Visual Grounding: Discovering Localization Heads Without Supervision*”
- Research Group: Vehicular Intelligence Lab

M. Tech. in Computer Science & Engineering

Aug 2019 – July 2021

Centre for Advanced Studies, Lucknow, India

B. Tech. in Computer Science & Engineering

Aug 2016 – July 2019

Azad Institute of Engineering & Technology, Lucknow, India

PUBLICATIONS

*Equal contribution †Corresponding author

Under Review

- N Sengar[†], A Saurez, D Har . “**CAGSeg: Training-Free Referring Segmentation via Contrastive Attention Grounding in Vision-Language Models .**” *IEEE Transactions on Circuits and Systems for Video Technology* .

Journal Articles

- N Sengar[†], A Saurez, D Har . “**Few Contrastive Attention Heads Enable Visual Grounding in Large Vision-Language Models .**” *Transactions on Machine Learning Research (TMLR)*, 2026 [J2C Certification] .
- N Sengar, Y Lee, K Won Ko, D Har, S Lee[†] . “**Vision-Based Tire Lifting for Parking by Dual Parking Robot .**” *IEEE Sensors Journal*, 2026 .
- N Sengar[†], I Kumari, J Lee, D Har . “**Blockchain-based Vision Transformer Network for Safe Driving Perception .**” *IEEE Sensors Journal*, 2024 . DOI
- S Mishra^{*†}, N Sengar^{*}, D Har . “**A Secure, Blockchain-Enabled Vehicular Sensor Communication Protocol With Deep Learning-Assisted Anomaly Detection .**” *IEEE Intelligent Transportation Systems Magazine*, 2024 . DOI
- Z Xu, N Sengar, T Chen[†], H Chung, O Oviedo-Trespalacios . “**Where is morality on wheels? Decoding large language model (LLM)-driven decision in the ethical dilemmas of autonomous vehicles .**” *Travel Behaviour and Society*, Elsevier, 2025 . DOI
- N Sengar, RC Joshi, MK Dutta[†], R Burget . “**EyeDeep-Net: A Multi-Class Diagnosis of Retinal Diseases using Deep Neural Network .**” *Neural Computing and Applications*, Springer, 2023 . DOI
- N Sengar, R Burget, MK Dutta[†] . “**A Vision Transformer Based Approach for Analysis of Plasmodium Vivax Life Cycle for Malaria Prediction Using Thin Blood Smear Microscopic Images .**” *Computer Methods and Programs in Biomedicine*, Elsevier, 2022 . DOI
- S Yadav, N Sengar, A Singh, A Singh, MK Dutta[†] . “**Identification of Disease using Deep Learning**

and Evaluation of Bacteriosis in Peach Leaf.” *Ecological Informatics*, 2021 . DOI

Conference Papers

- J Lee, Q Lai-Dang, N Sengar, D Har† . “Robust Monocular Depth Estimation in Adverse Weather Conditions by Unsupervised Domain Adaptation .” *ECAI 2024* . DOI
- Z Xu, N Sengar, T Chen† . “Evaluation of Ethical Dilemmas in ChatGPT-Based Decision-Making of Autonomous Vehicles .” *The 91st Conference of Korean Society of Transportation* . DOI
- N Sengar†, RC Joshi, MK Dutta . “An Efficient Artificial Intelligence-based approach for Diagnosis of Media Haze Disease .” *ICCCNT 2021, IEEE* . DOI
- N Sengar†, A Singh, V Yadav . “Classification of Documents using Bidirectional Long Short-Term Memory Recurrent Neural Network .” *ICSCSP20, Springer* . DOI
- N Sengar†, A Singh, S Yadav, MK Dutta . “Automated System for Face-mask Detection using Convolutional Neural Network .” *ICMC, Springer* . DOI
- N Sengar†, N Fatima, V Yadav . “Auto Organizer: A Machine Learning based Tool for Automatic Organization of Files .” *International Conference on Signal Data Processing* . DOI
- N Sengar†, T Pan, S Das, KB Singh, SC Misra . “Pneumocystis Jirovecii Fungi Detection Using Microscopy Image: A Deep Learning Approach .” *ICAIE 2023, Springer* . DOI
- A Malviya, N Sengar, MK Dutta†, R Burget, V Myska . “Deep Learning Based Gastro Intestinal Disease Analysis Using Wireless Capsule Endoscopy Images .” *45th International Conference on Telecommunications and Signal Processing (TSP 2022), IEEE* . DOI
- A Singh, N Sengar, MK Dutta†, R Burget . “Predicting Lung Cancer Through Histopathological Tissue Images .” *45th International Conference on Telecommunications and Signal Processing (TSP 2022), IEEE* . DOI

Patent

- S. C. Misra, D. Das, V. Udutalapally, S. Misra, N Sengar, A. Ghosh . “Microcontroller Blockchain Network with Edge Cloud Interplay for Data Security from Source .” *Patent No. 529261* .

RESEARCH EXPERIENCE

Graduate Research Assistant

Feb 2023 – Present

Vehicular Intelligence Lab, KAIST (Advisor: Prof. Dongsoo Har)

- My dissertation shows that frozen LVLMS already encode spatial grounding knowledge in a small subset of attention heads, discoverable through contrastive prompting without any labeled data or fine-tuning.
- **Referring Expression Comprehension:** Training-free attention-head discovery framework; outperforms training-free baselines by up to 27.95% on RefCOCO.
- **Referring Expression Segmentation:** Extended to pixel-level masks via spectral label propagation—competitive with supervised methods, no external segmentation model.
- **Visual Question Answering:** Generalizing contrastive head discovery to open-ended visual reasoning.
- **Parking Robot:** Developed a vision-based perception pipeline for a dual parking robot, including tire contact-point detection and camera-LiDAR steering angle estimation, achieving $\leq 2^\circ$ alignment error for autonomous alignment.
- Built a centralized ROS 2 fleet server coordinating multiple parking robots to lift and transport a vehicle as a rigid body, with synchronized lift/carry barriers and a shared-frame wheelbase safety check.
- **Charging Robot:** Developed a vision-based 6DOF pose estimation system for autonomous EV charging robot using object detection, PnP solving, and AprilTag-based localization.

Research Assistant (Part-time)

July 2024 – Oct 2024

Human Factor Centered Transport Safety Lab, KAIST (Advisor: Prof. Tiantian Chen)

- Investigated ethical decision-making of LLMs in autonomous vehicle dilemmas, evaluating how ChatGPT prioritizes pedestrians vs. passengers in trolley-problem scenarios.

Junior Research Fellow

June 2022 – Jan 2023

Indian Institute of Technology Kanpur (Advisor: Prof. Subhas Chandra Misra)

- Built a microcontroller blockchain network with edge-cloud interplay for secure agricultural data collection as part of the MEITY-funded "TribeConnect" smart tribal eco-platform project in Chhattisgarh.

Junior Research Fellow

Aug 2021 – May 2022

Centre for Advanced Studies (Advisor: Prof. Malay Kishore Dutta)

- Developed an AI-based assistive device for visually impaired users, generating real-time scene captions as part of a DST-funded research project.
- Developed a low-cost IoT-based intelligent stethoscope with embedded AI for early diagnosis of heart, lung, and prenatal conditions as part of a DST-funded project.

Artificial Intelligence Intern

May 2021 – Aug 2021

Farm365 Innovations (Startup)

- Developed a supervised learning-based neural network for automated animal intrusion detection and deterrence in agricultural fields.

KIT Winter Internship Program

Dec 2019 – Jan 2020

Indian Institute of Technology Kanpur (Advisor: Dr. Vipul Arora)

- Implemented a query-by-humming song retrieval system using audio feature matching, in collaboration with a PhD researcher.

AWARDS & HONORS

2026	Nominated as an Alumni Member of the Board of Studies	Centre for Advanced Studies
2025	Selected for HKUST Overseas Study Program for Entrepreneurship Bootcamp	HKUST
2023	KAIST Graduate Scholarship (full funding, 4 years)	KAIST
2019	Awarded State Government Scholarship for M.Tech (2 years)	Uttar Pradesh, India

TECHNICAL SKILLS

Programming	Python (advanced), Embedded C/C++, CUDA
ML / DL	Transformers, PyTorch, TensorFlow, Hugging Face Transformers
Computer Vision	OpenCV, 6DOF Pose Estimation, Open3D
Robotics	ROS 2, SLAM, Fast-DDS, multi-robot fleet coordination, LattePanda Sigma, Hailo-8 AI Processor, Raspberry Pi, Realsense D435i (depth camera), Sony IMX415, Livox Mid-70 LiDAR, ESP32 Microcontrollers, other utility sensors
Tools	Linux, L ^A T _E X, Arduino IDE, Edge Impulse, Roboflow

PROFESSIONAL SERVICE

Reviewer (Top-Tier Venues)

- Journals: IEEE T-ITS (2026), IEEE Sensors (2023-2026), IEEE IoT (2024)
- Conferences: ICML (2026), IJCAI (2026), NeurIPS (2025), IEEE Intelligent Vehicles Symposium (IV) (2024)

Volunteer

- Student Ambassador in Here Comes KAIST (HCK) program, 2025-26
- Student Coordinator in KAIST International Summer School (KISS), 2024
- Student Mentor for the new Fall International Students, 2023, 2025
- Student Volunteer, International Conference on Contemporary Computing and Application (IC3A2020)
- Student Volunteer, Dr. Abdul Kalam Technical, Literary and Management Fest 2019-20 (State Level)

INVITED TALKS & CONFERENCE PRESENTATIONS

2025	<i>Vision-Language Models for Scene Understanding in Autonomous Driving</i> (Invited Talk)	Workshop "AI in Robotics", Amity University
2024	<i>Evaluation of Ethical Dilemmas in ChatGPT-Based Decision-Making of Autonomous Vehicles</i>	91st Conference of the Korean Society of Transportation (KST), International Oral Session